

DEVSHREE JADEJA

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Education

Stony Brook University, United States

M.S. in Data Science

Relevant Coursework: Statistical coding in R, Data Analytics, Probability & Statistics

Aug 2024 – Expected May 2026

GPA: 3.89/4

Pandit Deendayal Energy University, India

B.Tech. in Computer Science

Relevant Coursework: Deep Learning, Database Systems, NLP, CV

Oct 2020 – May 2024

GPA: 3.96/4

Professional Experience

iLocalBox LLC, CA, USA

Machine Learning Intern

May 2025 – Aug 2025

- Designed **GNN-based recommenders** by benchmarking **Face Transformers** on 2K+ user-item graphs for neighbor-aware ranking; Raised accuracy by 17% and session ROI by 13% in NoSQL databases.
- Optimized transformer-based generative models for real-time UI/voice/text adaptation; reduced cross-device latency by 35ms and Augmented **multilingual query alignment** in GenAI-assisted creative flows.
- Assessed **multi-label VLM classifiers** with class-balanced loss for predicting churn, toxicity, and satisfaction across 1M+ user traces; integrated **SHAP** and **Grad-CAM** to enhance interpretability; advanced F1 by 24%.
- Automated PySpark + MLflow for **online inference**, cutting XML Data problems 4× and ranking lag to 5min.

ISRO, Gujarat, India

Research Intern

Dec 2023 – Aug 2024

- Improved **CTR/CVR prediction** via **Siamese BERT with JAX** for query-ad semantic matching, **Random Forest** for classification, and Transformer seq2seq for query understanding with SQL; achieved +22% AUC and higher tail recall.
- Tuned **multimodal deep learning pipelines** (text/image/video) in TensorFlow, PyTorch, and Keras for real-time **relevance modeling** and CUDA programming for GPU acceleration; reduced false positives by 27%.
- Scaled **Kafka-based ETL workflows** and CI/CD pipelines (Spark, Hadoop, Docker) for ads budget pacing with MSSQL, delivery throttling, **ranking diagnostics**; cut deployment latency by 60%.
- Applied **feature engineering** and **A/B testing** to optimize ad delivery; delivered 12% lift in performance.

Academic Projects

ShopSmart – Price Optimization Engine

Jan 2025 - May 2025

- Initiated **XGBoost/LightGBM** for marketplace **price prediction**, boosting conversions 12% with **Optuna & SHAP**.
- Crafted **NLP pipeline** for sentiment analysis, query understanding improving **user modeling** & personalization.
- Launched scalable **FastAPI service** with Streamlit/Looker for real-time pricing insights; cut manual reporting by 60%.

TemporalPlay – Player Modeling & Intent Forecasting in AR/VR Games

Jun 2024 – Aug 2024

- Built **intent classification & churn models** (RLHF + CatBoost), improving F1 by **26%**.
- Designed **attention-based temporal models in PyTorch**, with **contextual bandits RL** for decision-making.
- Developed **closed-loop personalization pipelines** with contextual action embeddings to optimize player engagement.

SpatialScene3D – Multimodal 3D Scene Understanding

Apr 2024 – Jun 2024

- Refined a 3D object detection pipeline using 2D segmentation, ray casting, and depth fusion with 92% consistency.
- Deployed **LLM-based RAG Retrieval** on 3D scenes, achieving 94% answer accuracy across 500+ spatial queries.

Publications

- Real-Time Lane Detection for Autonomous Driving & GRAD-CAM Visualization; *Springer*, [Link].
- From Speech-to-Spatial: Grounding Utterances on a Live Shared View with Augmented Reality; *IEEE VR* [Link].
- SpeechLess: Micro-utterance with Personalized Spatial Memory-aware Assistant; *IEEE VR* [Link].

Skills

Programming & Scripting: Python, SQL, Java, Scala, JavaScript, C, C++, Bash, Git

distributed training framework: DDP, FSDP, Deepspeed, Megatron, or HuggingFace Accelerate, Ray

Machine Learning & AI: TensorFlow, PyTorch, Transformers, LLMs, RAG, Deep Learning, NLP, Computer Vision

Cloud: AWS, Docker, Airflow,3, Glue, Athena, SageMaker, Bedrock, Lambda, and OpenSearch, GCP, Spark, Kubernetes